

pICkIT: Automated Analysis of Protein–Ligand Interactions

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Abstract

pICkIT is an open-source Python library designed to automate the extraction, filtering, and analysis of protein–ligand interaction data from large structural datasets. The tool was developed to address the increasing need for efficient, scalable processing of interaction files generated by tools such as Arpeggio, whose datasets are often too large and complex for manual inspection. pICkIT provides a modular framework that integrates interaction parsing, automated summarization, dataset comparison, and publication-ready analysis, enabling researchers to rapidly identify meaningful interaction patterns across hundreds or thousands of complexes. In its validation study, pICkIT processed hundreds of protein–ligand complexes, reducing hundreds of megabytes of raw interaction output to compact summary files while preserving key structural information. The library facilitates high-level tasks such as filtering by interaction type, quantifying residue-level contact frequencies, generating customizable heatmaps, and comparing interaction profiles across ligand series. By simplifying these steps, pICkIT accelerates workflows, QSAR model development, and structure-based drug design.

What does pICkIT do?

- Parses large Arpeggio JSON interaction files
- Filters interactions by type, residue, or region
- Integrates ligand activity (pIC50) data
- Generates compact summary matrices (.xlsx)
- Produces automatic plots and heatmaps

Workflow: from PDB to interaction insights

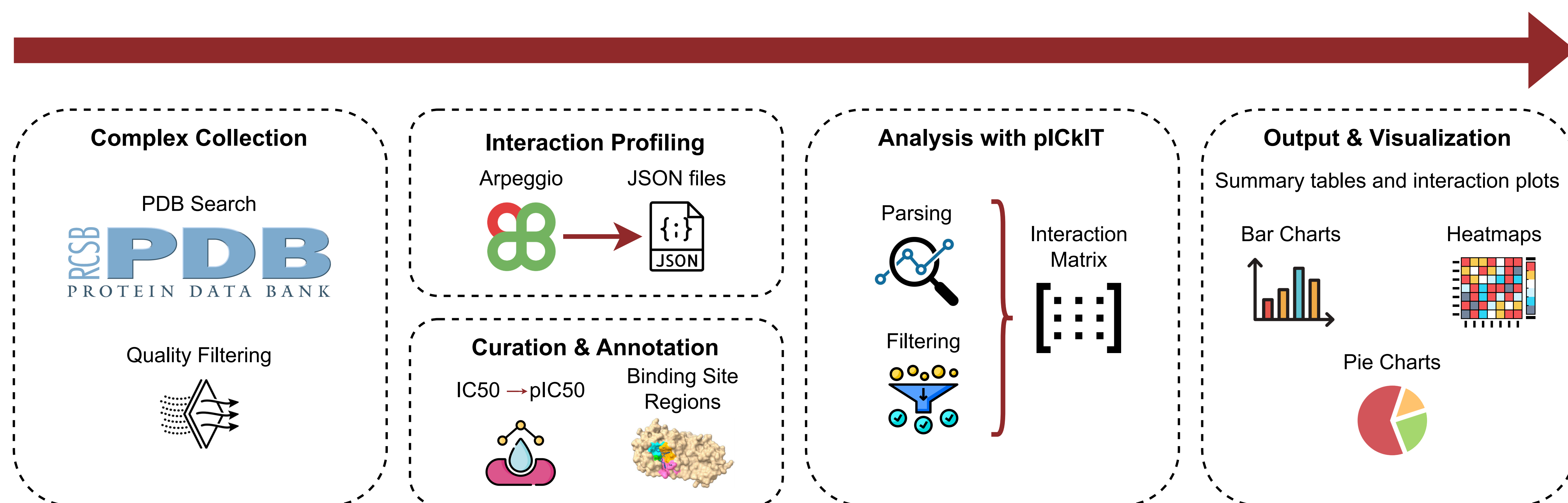


Figure 1: pICkIT workflow. Pipeline from protein–ligand complex collection and Arpeggio interaction extraction to data curation, pICkIT processing, and result plots.

Application in diverse SARS-CoV-2 M_{PRO} inhibitors

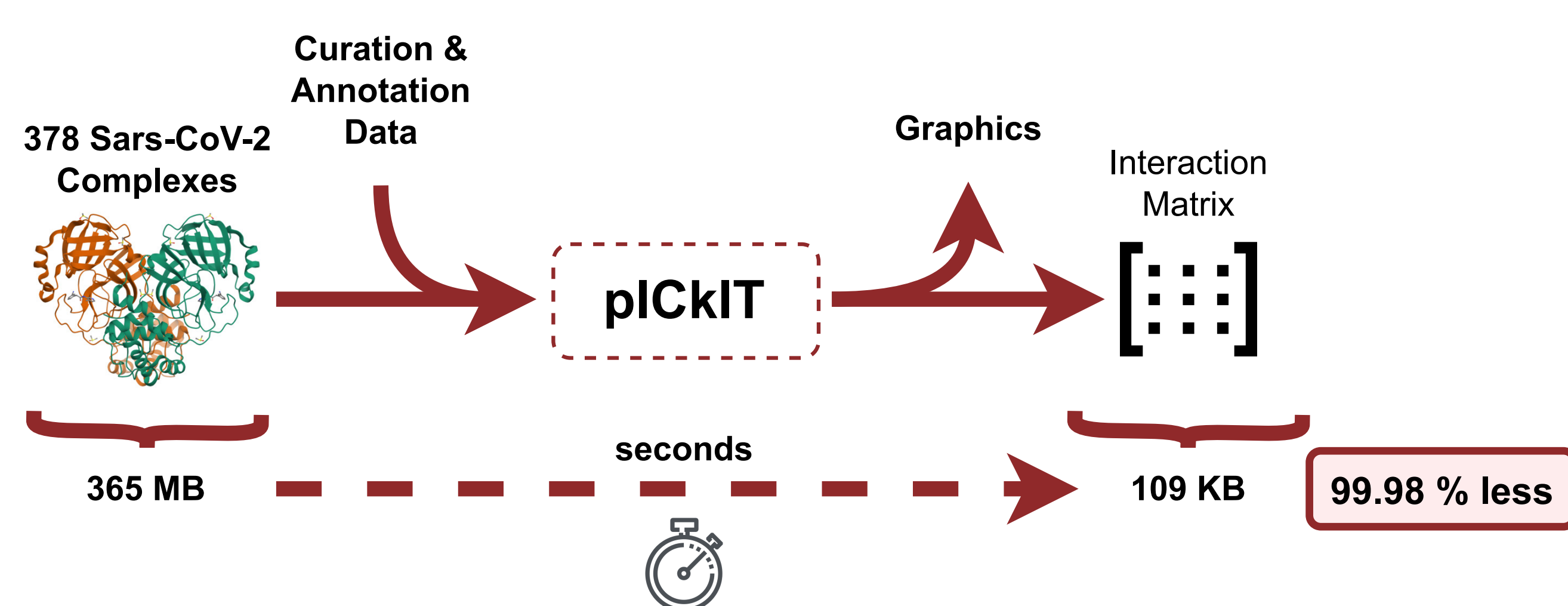


Figure 2: Application to SARS-CoV-2 M_{PRO} complexes. Real-case workflow highlighting the 99.98% data reduction achieved by pICkIT and analysis performed in seconds.

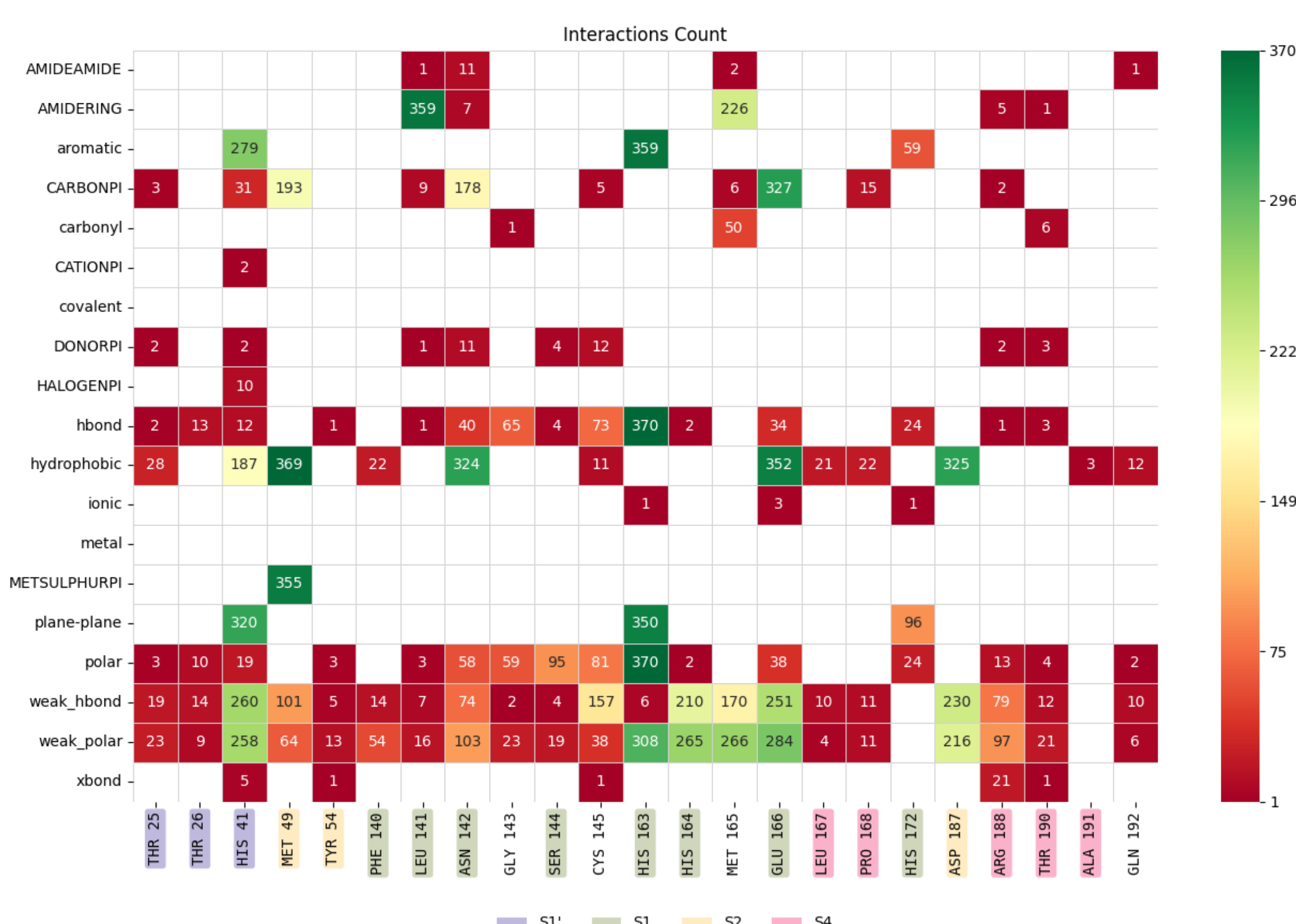


Figure 3: Interaction frequency heatmap. Counts of residue–interaction occurrences across the analysed complexes.

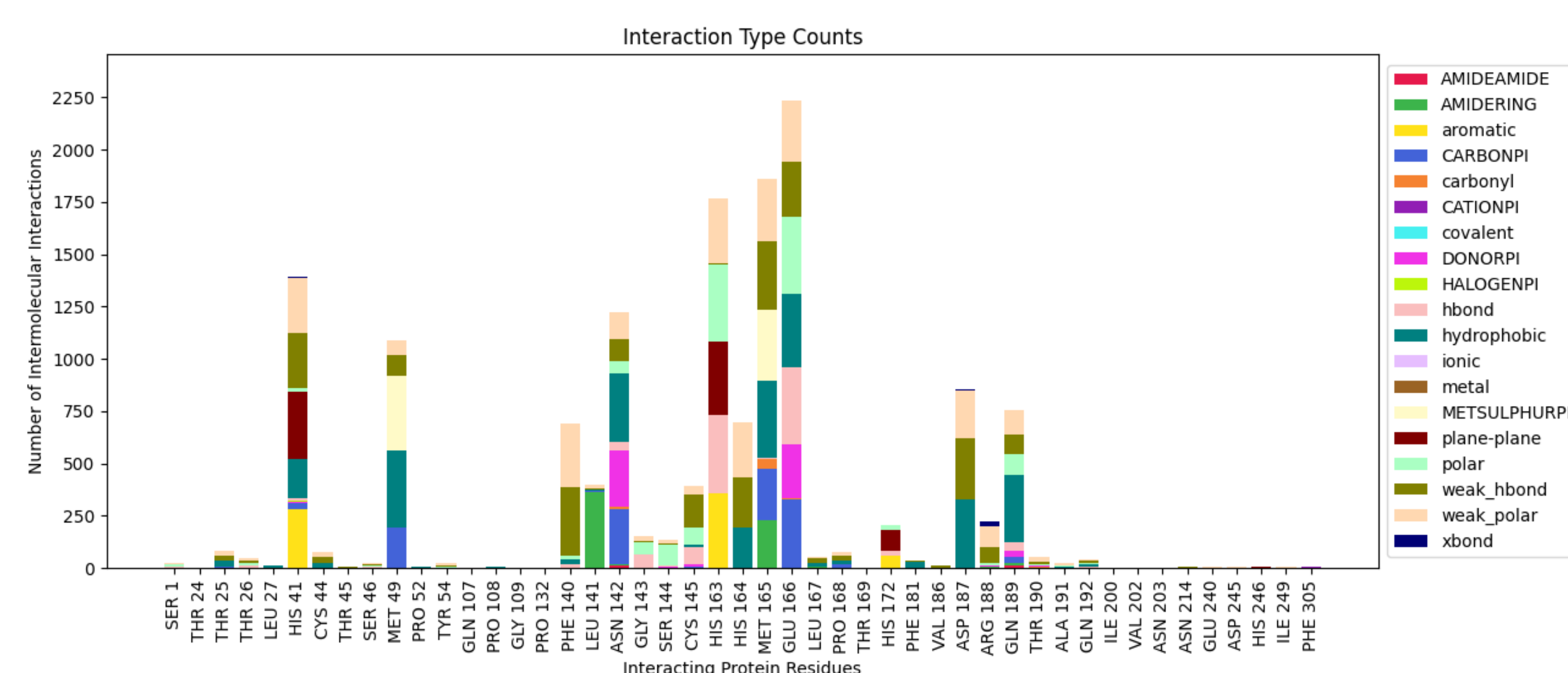


Figure 4: Residue interaction profile. Stacked bar chart showing the most active residues and predominant interaction types.

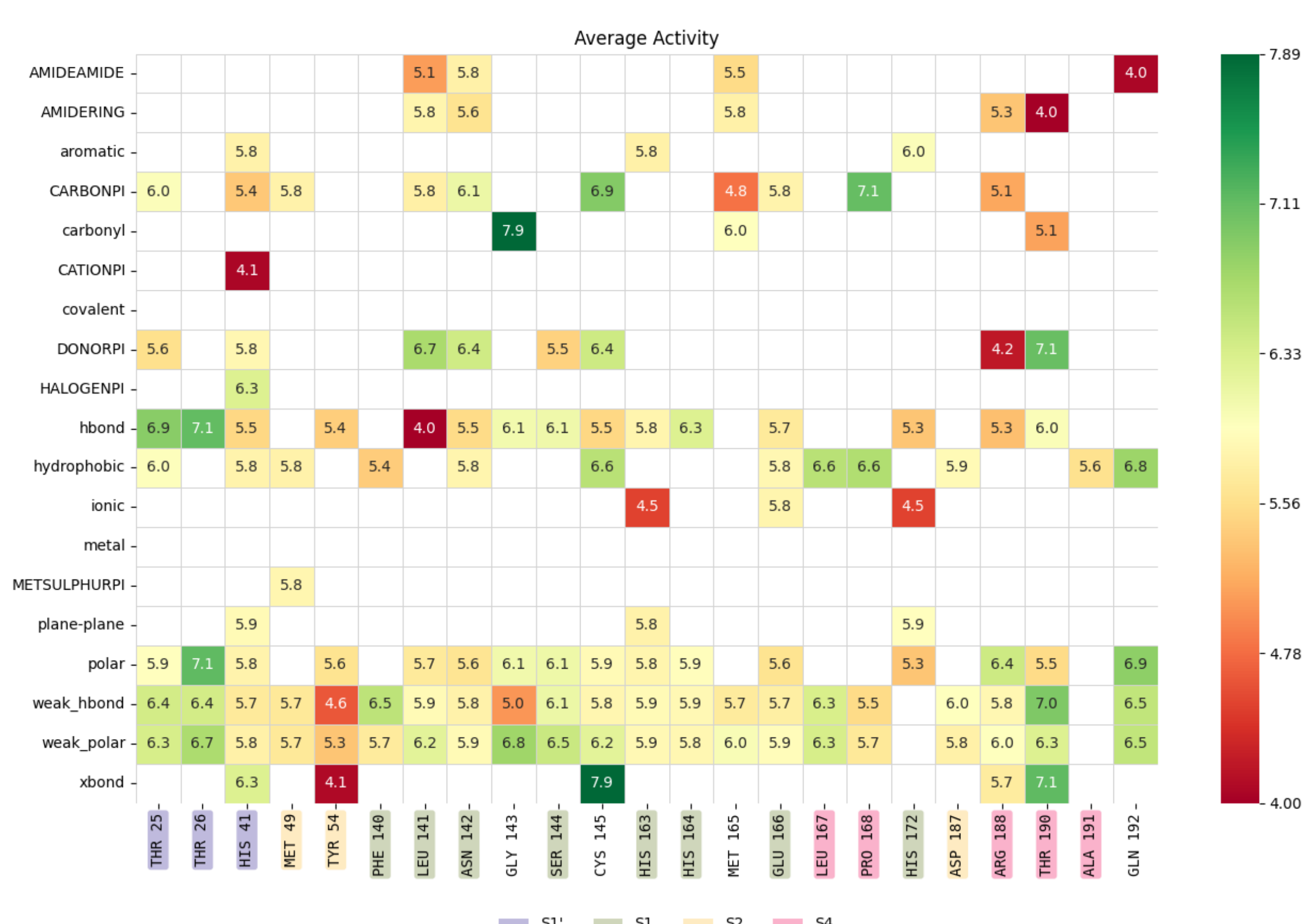


Figure 5: Mean activity heatmap. Average activity of complexes exhibiting each residue–interaction pair.